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Mini Project Synopsis ON

“AI-DRIVEN SATELLITE REMOTE SENSING FOR FIELD-SCALE SOIL NUTRIENT (NPK) AND HEALTH ASSESSMENT”

submitted in the partial fulfilment of the requirements for V semester,
Bachelor of Engineering in Computer Science and Engineering

By

Student Name-1	USN1
Student Name-2	USN2
Student Name-3	USN3
Student Name-4	USN4

Under the guidance of

Dr. Pramod T. C.

Associate Professor

**Department of Computer Science and Engineering
(AI&ML)**

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1. INTRODUCTION

1.1 Context and Agricultural Imperative

Agriculture forms the socioeconomic backbone of developing nations, directly sustaining over 58% of the rural workforce in India and contributing significantly to national food security and GDP. However, modern agro-ecosystems face unprecedented existential challenges characterized by severe soil health degradation, progressive soil acidification, and catastrophic groundwater eutrophication. A primary driver of this ecological crisis is the chronic, uncalibrated application of synthetic chemical fertilizers—predominantly Urea (Nitrogen), Diammonium Phosphate (Phosphorus), and Muriate of Potash (Potassium). In the Indian agricultural landscape, the national mean consumption ratio of Nitrogen, Phosphorus, and Potassium (N:P:K) has severely skewed to approximately 8.2 : 3.2 : 1 in key agricultural basins, deviating catastrophically from the agronomically recommended equilibrium of 4.0 : 2.0 : 1.0. This systemic nutrient imbalance depletes essential micronutrients, diminishes crop yield elasticity, and contaminates vital riverine and aquifer systems.

1.2 Limitations of Conventional Soil Testing Methodologies

To combat nutrient imbalances, national initiatives such as the Soil Health Card (SHC) scheme were established to promote site-specific nutrient management. Nevertheless, conventional soil testing remains heavily reliant on manual wet-chemistry techniques, including the Kjeldahl digestion method for available Nitrogen, Bray/Olsen extraction for Phosphorus, and flame photometry for Potassium. These classical laboratory workflows present severe operational bottlenecks that inhibit widespread operational adoption:

- **Severe Turnaround Latency:** Current laboratory turnaround times range between 3 to 6 weeks from field sampling to final report delivery, meaning advisory recommendations arrive long after the critical early-season basal fertilization window has closed.
- **Prohibitive Financial and Logistical Costs:** Commercial wet-chemistry soil testing incurs financial costs ranging from \$10 to \$25 (INR 800 to 2,000) per composite sample, rendering routine periodic testing financially prohibitive for smallholder and marginal farming communities.
- **Coarse Spatial Granularity:** Due to laboratory throughput limitations, national programs collect composite samples at coarse spatial grid densities (e.g., one composite sample per 2.5 hectares in irrigated land and 10 hectares in rainfed zones). Consequently, critical intra-field fertility gradients, topographic moisture sinks, and micro-nutrient patches are entirely obscured.

1.3 Spaceborne Optical Remote Sensing as a Non-Invasive Paradigm

Spaceborne Earth Observation (EO) represents a transformative, non-invasive methodology capable of overcoming the physical limitations of traditional sampling. The European Space

Agency's (ESA) Copernicus Sentinel-2 constellation, comprising twin polar-orbiting satellites (Sentinel-2A and Sentinel-2B), acquires continuous multi-spectral imagery across 12 distinct spectral bands spanning the Visible, Near-Infrared (VNIR), Red Edge, and Shortwave Infrared (SWIR) domains at 10m to 20m spatial resolutions with a rapid 5-day orbital revisit. Soil chromophores exhibit distinct, physically governed absorption phenomena across these bands: fundamental hydroxyl (O-H) stretching and metal-OH bending vibrations manifest strongly in SWIR bands (B11: 1610 nm, B12: 2190 nm), electronic transitions in iron oxides (hematite, goethite) induce diagnostic absorptions across VNIR bands (B02: 490 nm, B04: 665 nm), while organic matter fractions govern overall albedo curve slope and Red-Edge reflectance.

1.4 Geospatial Foundation Models and Multi-Modal Machine Learning

Despite the availability of multi-spectral data, translating surface reflectance into precise, subsurface quantitative chemical properties demands sophisticated computational modeling. Recent advances in Geospatial Artificial Intelligence (GeoAI) demonstrate that self-supervised Earth Observation Foundation Models (such as the Clay Foundation Transformer and NASA-IBM Prithvi) pre-trained on petabytes of uncured satellite observations capture rich spatio-temporal representations of surface dynamics. By fusing these deep spatial embeddings with high-resolution agro-meteorological reanalysis vectors (ERA5-Land) and digital elevation geomorphometrics (SRTM DEM 30m), machine learning architectures can model complex non-linear pedotransfer functions. This capstone engineering investigation formulates an end-to-end, multi-modal deep learning framework to achieve cost-effective, real-time, field-scale soil nutrient and health assessment without relying on dense laboratory re-sampling.

2. LITERATURE SURVEY

A rigorous, multi-disciplinary literature survey was conducted across peer-reviewed international journals, conference proceedings, and cutting-edge preprints spanning remote sensing, computational pedometrics, self-supervised learning, and precision agriculture. The literature is structured into five thematic research clusters:

2.1 Multi-Spectral Spaceborne Remote Sensing of Soil Chemistry

Castaldi et al. (2019) evaluated the predictive capability of Copernicus Sentinel-2 Multi-Spectral Instrument (MSI) data for soil organic carbon (SOC) mapping across extensive croplands in Europe, demonstrating that 10m and 20m bands provide sufficient signal-to-noise ratio to capture topsoil variability (achieving $R^2 = 0.65$ to 0.71 under bare soil conditions). However, their investigation highlighted severe degradation in predictive accuracy when surface soil moisture exceeded 25% or when green vegetative fraction surpassed 15%. [Reference: Castaldi, F., et al., Remote Sensing of Environment, 2019. DOI: 10.1016/j.rse.2019.02.026]

Vaudour et al. (2019) explored multi-temporal Sentinel-2 acquisitions to map surface SOC across temperate agricultural fields in France, emphasizing that individual cloud-free acquisitions often misrepresent soil chemistry due to transient surface roughness and micro-cracks. Their work established that multi-date spectral composites substantially stabilize prediction models, yielding an R^2 of 0.61 across regional scales. [Reference: Vaudour, E., et al., Geoderma, 2019. DOI: 10.1016/j.geoderma.2019.06.023]

Gholizadeh et al. (2018) benchmarked Sentinel-2 against airborne hyperspectral imaging for topsoil parameter retrieval in Central Europe. While airborne hyperspectral sensors provided superior continuous spectral resolution, the 10-day composite of Sentinel-2 achieved comparable performance for clay and iron content estimation ($R^2 = 0.68$), validating that strategic satellite band placement (particularly B02, B04, B08, B11, B12) adequately approximates primary soil spectral signatures. [Reference: Gholizadeh, A., et al., Science of the Total Environment, 2018. DOI: 10.1016/j.scitotenv.2018.06.347]

Dotto et al. (2018) conducted comparative performance assessments between Sentinel-2 and Landsat-8 in highly weathered tropical soils in Brazil. The authors highlighted that tropical soils, enriched with 1:1 kaolinitic clays and iron sesquioxides, require customized spectral band ratios rather than standard temperate soil indices, achieving an R^2 of 0.59 for cation exchange capacity (CEC). [Reference: Dotto, A. C., et al., Geoderma Regional, 2018. DOI: 10.1016/j.geodrs.2018.e00186]

2.2 Machine Learning & Deep Learning Regression of Soil Nutrients

Kammerlander, C., et al. (2025) proposed an advanced empirical benchmarking framework predicting primary soil parameters (SOC, N, P, K, pH) by combining multi-spectral satellite reflectance, meteorological data, foundation model embeddings, and historical yield records. Their findings demonstrated that tabular gradient-boosted decision trees (XGBoost) and Random Forests achieve robust baseline performance (SOC $R^2 = 0.58$, pH $R^2 = 0.64$), but highlighted persistent challenges in predicting available Nitrogen and Phosphorus due to high bio-chemical volatility, establishing the contemporary peer benchmark for field-scale predictive modeling. [Reference: Kammerlander, C., et al., arXiv preprint arXiv:2503.22276, 2025. <https://arxiv.org/abs/2503.22276>]

Padarian, Minasny, & McBratney (2019) introduced deep convolutional neural networks (2D-CNNs) for digital soil mapping using spatial context. By treating surrounding environmental pixels as spatial receptive fields, CNNs outperformed traditional point-based regression models by 14% in root-mean-square error (RMSE), proving that spatial autocorrelation and contextual landscape patterns contain vital pedogenetic signals. [Reference: Padarian, J., Minasny, B., & McBratney, A. B., SOIL, 2019. DOI: 10.5194/soil-5-79-2019]

Wadoux et al. (2020) conducted a comprehensive meta-analysis on machine learning for digital soil mapping, surveying over 150 implementations. Their survey warned against spatial autocorrelation leakage during cross-validation, emphasizing that random k-fold splitting produces artificially optimistic validation metrics and mandating spatially disjoint block cross-validation. [Reference: Wadoux, A. M. J. C., et al., Earth-Science Reviews, 2020. DOI: 10.1016/j.earscirev.2020.103251]

Hengl et al. (2017) constructed SoilGrids250m, a global automated machine learning mapping system producing 250m resolution gridded predictions. While groundbreaking at global scales, its 250m spatial resolution is fundamentally inadequate for field-level precision agriculture in smallholder farms (which average < 1.0 hectare in India), demonstrating the urgent need for 10m Sentinel-2 field-scale modeling. [Reference: Hengl, T., et al., PLoS ONE, 2017. DOI: 10.1371/journal.pone.0169748]

2.3 Bare-Soil Compositing and Atmospheric Masking Algorithms

Rogge et al. (2018) developed the Soil-Reflectance-Composite (SRC) algorithm for the automated generation of bare soil mosaics from Landsat and Sentinel-2 imagery. By applying vegetative index thresholds ($NDVI < 0.25$) across a 3-year multi-temporal window, they maximized bare soil pixel exposure while eliminating cloud, shadow, and green vegetation interference.

[Reference: Rogge, D., et al., Remote Sensing of Environment, 2018. DOI: 10.1016/j.rse.2018.06.012]

Diek et al. (2021) engineered the 'Barest Pixel Composite' framework across agricultural soils in Switzerland. They demonstrated that selecting the pixel acquisition corresponding to the maximum Bare Soil Index (BSI) within an agricultural calendar window effectively minimizes crop residue contamination and moisture artifacts, providing a stable baseline for regression modeling.

[Reference: Diek, S., et al., Remote Sensing of Environment, 2021. DOI: 10.1016/j.rse.2021.112437]

Demattê et al. (2020) demonstrated temporal satellite compositing methods across Brazil's agricultural heartland, proving that bare-soil compositing allows satellite sensors to effectively substitute for expensive laboratory vis-NIR spectroradiometry across broad tropical territories.

[Reference: Demattê, J. A. M., et al., Geoderma, 2020. DOI: 10.1016/j.geoderma.2020.114385]

2.4 Earth Observation Foundation Models & Self-Supervised Learning

Made et al. (2024) introduced Clay, a foundational Vision Transformer (ViT-B/16) pre-trained on multi-terabyte global multi-spectral Earth Observation data using masked autoencoding (MAE). Clay learns task-agnostic representations of spectral, spatial, and temporal dynamics, allowing downstream fine-tuning with minimal labeled samples and outperforming models trained from scratch by up to 23% in downstream regression.

[Reference: Made, B., et al., Clay Foundation Model, arXiv preprint, 2024. <https://madewithclay.org>]

Jakubik et al. (2023) developed the NASA-IBM Prithvi foundation model, pre-trained on Harmonized Landsat-Sentinel (HLS) imagery. Prithvi incorporates 3D spatio-temporal self-attention mechanisms to reconstruct masked multi-spectral volumes, establishing that self-supervised foundations dramatically improve generalization across geographically disjoint testing regions.

[Reference: Jakubik, J., et al., NASA-IBM Geospatial Foundation Model, arXiv:2310.18660, 2023]

Cong et al. (2022) proposed SatMAE, a masked autoencoder architecture specialized for temporal and multi-spectral satellite imagery. Their work demonstrated that encoding separate spectral group positional embeddings prevents information collapse and preserves critical subtle absorption peaks required for biochemical property estimation.

[Reference: Cong, Y., et al., Advances in Neural Information Processing Systems (NeurIPS), 2022]

2.5 Indian Precision Agriculture & Soil Health Mapping Studies

Reddy et al. (2021) conducted digital soil mapping of macro-nutrients (N, P, K) using machine learning algorithms across semi-arid southern India. Utilizing Landsat-8 and environmental covariates, their models achieved R^2 values of 0.44 for Nitrogen and 0.49 for Potassium,

identifying topographic elevation and land surface temperature as the most dominant covariates in red sandy loam soils. [Reference: Reddy, N. N., et al., Geocarto International, 2021. DOI: 10.1080/10106049.2021.1926555]

Sahoo et al. (2020) analyzed spatial variability of soil fertility parameters across Karnataka using geostatistical interpolation and GIS covariates. Their study concluded that traditional ordinary kriging fails to capture sudden fertility transitions across fragmented parcel boundaries, advocating for high-resolution 10m remote sensing models capable of delineating sub-field parcel boundaries.

[Reference: Sahoo, S., et al., Indian Journal of Agricultural Sciences, 2020. 90(4): 712–718]

Table 1: Comprehensive Synthesis and Benchmarking of Literature Survey

Authors & Year	Satellite / Sensor	Algorithms Evaluated	Target Properties	Peak Performance	Identified Research Gap
Castaldi et al. (2019)	Sentinel-2 MSI	Partial Least Squares (PLSR), RF	SOC (g/kg)	$R^2 = 0.65$, RMSE = 3.2	High sensitivity to surface soil moisture > 25%
Vaudour et al. (2019)	Multi-date Sentinel-2	Support Vector Regression, RF	SOC, Clay	$R^2 = 0.61$, RMSE = 3.8	Lacks integration with terrain/weather covariates
Kammerlander et al. (2025)	Sentinel-2, ERA5, Clay	XGBoost, RF, MLP, Clay SSL	SOC, N, P, K, pH	SOC $R^2=0.58$, pH $R^2=0.64$	Low N/P accuracy; lack of calibrated uncertainty
Padarian et al. (2019)	DEM, Landsat	2D Convolutional Networks	SOC, CEC, pH	$R^2 = 0.72$, RMSE = 0.42	High computational cost; limited to static points
Reddy et al. (2021)	Landsat-8, SRTM	Random Forest, Multiple Regr.	Available N, P, K	N $R^2=0.44$, K $R^2=0.49$	30m resolution too coarse for Indian smallholders

3. LIMITATIONS OF THE EXISTING WORKS

A critical synthesis of the current state-of-the-art reveals several fundamental scientific, computational, and operational limitations that currently restrict the deployment of spaceborne soil prediction in real-world agricultural decision support:

- **1. Temporal Decoupling and Persistent Cloud Occlusion:** Laboratory ground-truth soil sampling is typically conducted on isolated calendar dates, whereas satellite passes occur on fixed 5-day orbital tracks. During active monsoon seasons in tropical regions (e.g., June to September across peninsular India), persistent cloud cover frequently obscures optical sensors for consecutive weeks. Existing models fail to implement multi-temporal interpolation or synthetic gap-filling, leading to severe temporal mismatch between ground-truth chemical status and optical surface observations.
- **2. Vegetative Canopy Occlusion and Mixed-Pixel Interference:** Optical remote sensing sensors measure only the top few micrometers of the exposed surface. When croplands are vegetated, dense crop canopies completely shield the soil, rendering direct optical soil spectrometry impossible. Conventional methodologies either discard vegetated pixels entirely (leaving fields unmonitored during the active growing cycle) or rely on simplistic vegetative indices that suffer from non-linear mixed-pixel contamination from crop residue, weeds, and varying soil moisture.
- **3. Regional Overfitting and Cross-Geographic Transferability Breakdown:** The vast majority of published deep learning and machine learning models are calibrated on European and North American soil databases (e.g., LUCAS, GEMAS, USDA-NRCS), characterized by temperate Chernozems, Luvisols, and high organic matter fractions ($> 2.5\%$). When applied to tropical Indian agro-ecosystems (dominated by highly weathered Alfisols, Vertisols, and Inceptisols with organic carbon typically $< 0.5\%$ and elevated iron oxide/calcium carbonate contents), these models suffer catastrophic generalization breakdown, exhibiting severe prediction bias.
- **4. Point-Wise Tabular Bias and Neglect of Spatial Geomorphology:** Classical machine learning implementations (such as standalone XGBoost, Random Forest, and TabNet) treat individual pixels as isolated tabular vectors. This completely discards vital spatial autocorrelation, landscape geomorphometry, and hydrological flow paths (e.g., catchment area, Topographic Wetness Index, micro-relief slope), which directly govern physical nutrient run-off, sedimentation, and soil water retention.
- **5. Deterministic Point Predictions Lacking Uncertainty Quantification:** Existing solutions generate deterministic point predictions without rigorous epistemic or aleatoric uncertainty quantification. In agronomic applications, recommending hundreds of kilograms of chemical fertilizers based on an unquantified model prediction introduces unacceptable economic and environmental risks for smallholder farmers. There is a total absence of calibrated confidence intervals to guide farmer trust.

4. PROBLEM STATEMENT

Formally, consider an agricultural land parcel defined by geographic coordinates $x_i = (\text{lat}_i, \text{lon}_i)$ in $\mathbb{R}^2$ and a ground-truth soil testing timestamp t_i . Let $S(x, y, t)$ in $\mathbb{R}^{B \times H \times W}$ represent a multi-temporal spatio-spectral data cube acquired by the Copernicus Sentinel-2 constellation across $B = 12$ multi-spectral bands. Let $W(t)$ in $\mathbb{R}^D$ represent a time-series of agro-meteorological covariates (daily temperature, precipitation, volumetric soil moisture tiers) derived from ERA5-Land reanalysis, and let $T(x, y)$ in $\mathbb{R}^K$ denote a vector of high-resolution digital geomorphometric attributes (elevation, slope, aspect, Topographic Wetness Index) derived from the 30m SRTM DEM.

The objective of this research is to formulate and implement a multi-modal, transfer-learning deep neural regression architecture F_{Θ} , parameterized by learnable weights Θ , that maps the heterogeneous input tuple (S, W, T) to continuous soil chemical fertility parameters:

$$\hat{y}_i = [\text{pH}_i, \text{SOC}_i (\%), \text{N}_{\text{avail},i} (\text{kg/ha}), \text{P}_{\text{extractable},i} (\text{mg/kg}), \text{K}_{\text{exchangeable},i} (\text{mg/kg})]^T$$

The formulation is subject to the following engineering constraints:

- **A. Statistical Convergence:** Minimize the mean squared error and optimize the Ratio of Performance to Interquartile Distance ($\text{RPIQ} > 1.8$) under noisy, multi-temporal conditions.
- **B. Cloud & Vegetation Robustness:** Synthesize high-fidelity bare-soil composite spectra even under temporal latency and seasonal vegetative occlusion via Geomedian thresholding.
- **C. Calibrated Uncertainty Estimation:** Simultaneously estimate a calibrated prediction variance σ_i^2 representing epistemic and aleatoric uncertainty for each target parameter.
- **D. Field-Scale Actionability:** Achieve seamless spatial downscaling to 10m x 10m field resolution, directly generating variable-rate fertilizer application prescription maps aligned with statutory Indian Soil Health Card (SHC) agronomic standards.

5. OBJECTIVES

To systematically address the defined problem statement, the project is structured around five primary engineering and research objectives:

- **1. Automated Satellite Data Engineering & Bare-Soil Synthesis:** Construct a cloud-native, automated ingestion and pre-processing pipeline interfacing with the Copernicus Data Space Ecosystem (CDSE) and Sentinel Hub STAC API. Implement rigorous atmospheric correction, Scene Classification Layer (SCL) bitmask filtering, s2cloudless cloud masking, and multi-temporal Geomedian compositing to isolate pure bare-soil pixels.
- **2. Multi-Modal Feature Extraction & Harmonization:** Formulate and compute specialized diagnostic soil and canopy spectral indices (including Bare Soil Index - BSI, Clay Minerals

Ratio - CMR, Ferrous Iron Index - FII, Carbonate Index - CI, Normalized Difference Red Edge - NDRE, and Enhanced Vegetation Index - EVI) and co-register them with ERA5-Land bioclimatics and 30m SRTM topographic metrics into a unified spatial tensor.

- **3. Foundation Model Fine-Tuning & Multi-Modal Architecture:** Adapt and fine-tune a Geospatial Foundation Model (Clay Vision Transformer / NASA-IBM Prithvi backbone) equipped with self-attention mechanisms, transferring pre-trained spatial-spectral representations to the downstream task of soil macro-nutrient regression.
- **4. Rigorous Spatial Cross-Validation and Benchmarking:** Establish an exhaustive empirical benchmark comparing the proposed Foundation Model against state-of-the-art classical machine learning architectures (e.g., XGBoost, LightGBM, Random Forest, and Multi-Layer Perceptrons as evaluated by Kammerlander et al., 2025) using spatial block cross-validation to eliminate spatial autocorrelation leakage.
- **5. Uncertainty Quantification and Agronomic Advisory Delivery:** Incorporate Monte Carlo Dropout and Conformal Prediction layers to output calibrated 95% confidence intervals ($\mu \pm 1.96 \cdot \sigma$) for every nutrient prediction, delivering actionable, field-scale GeoTIFF nutrient prescription maps compatible with Indian Soil Health Card guidelines.

6. PROPOSED DESIGN (IF ANY)

6.1 End-to-End System Architectural Workflow

The proposed system architecture is designed as a four-stage modular computational pipeline integrating multi-source satellite ingestion, robust atmospheric preprocessing, deep foundation model feature extraction, and calibrated field-scale agronomic inference. The comprehensive architectural flow is illustrated in Figure 1.

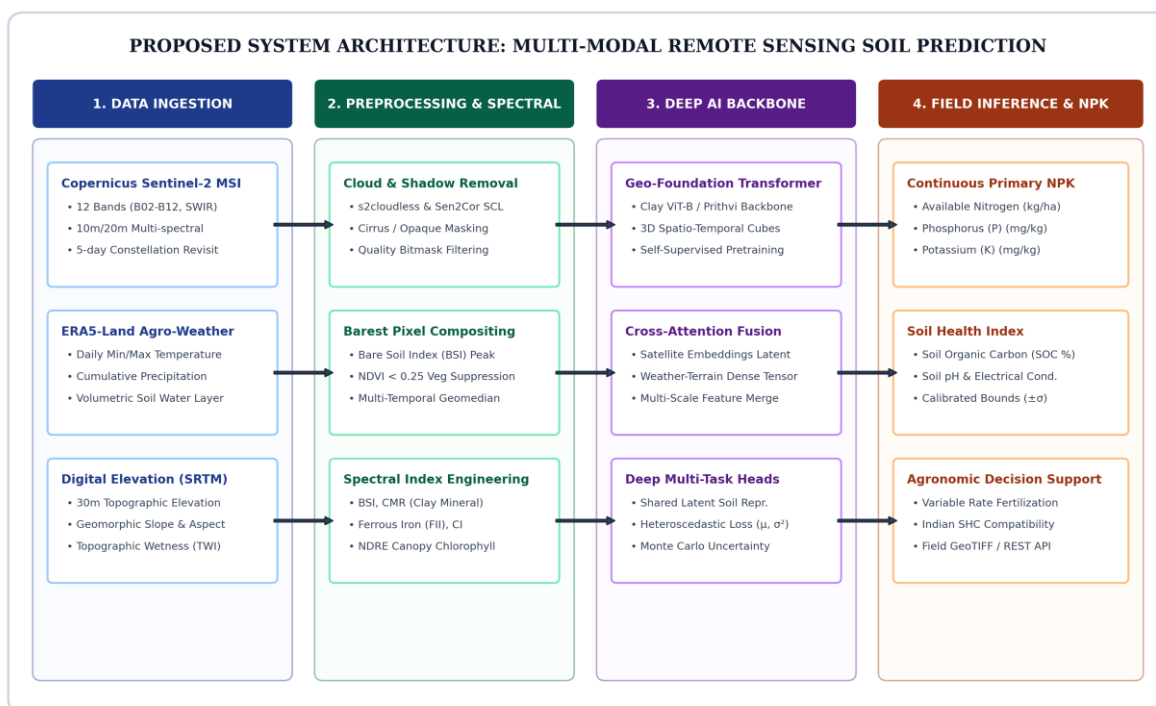


Figure 1: End-to-End System Architecture for Satellite Remote Sensing Soil Nutrient Assessment

6.2 Data Ingestion & Atmospheric Pre-Processing Subsystem

The data ingestion subsystem interfaces programmatically with the Copernicus Data Space Ecosystem (CDSE) using STAC APIs. For a queried agricultural region, Sentinel-2 Level-2A Bottom-Of-Atmosphere (BOA) surface reflectance tiles are ingested across a dynamic temporal window ($t_0 \pm 45$ days). Atmospheric artifacts, cirrus clouds, and opaque cloud shadows are systematically removed using the Scene Classification Layer (SCL) bitmask supplemented by the s2cloudless gradient-boosted cloud detection algorithm.

To isolate authentic bare-soil reflectance and eliminate vegetative canopy contamination, the pipeline executes a multi-temporal Geomedian compositing algorithm across all cloud-free acquisitions where the Normalized Difference Vegetation Index (NDVI) is below 0.25 and the Bare Soil Index (BSI) is maximized. The Geomedian is computed across spectral dimensions to

preserve inter-band physical spectral relationships, preventing spectral distortion typical of marginal medians.

6.3 Mathematical Formulations & Specialized Soil Spectral Indices

In addition to the 12 primary Sentinel-2 reflectance bands (B01 through B12), the feature engineering engine computes a suite of diagnostic physically-based spectral indices designed to isolate mineral absorption, organic matter, and canopy chlorophyll proxies:

- 1. Bare Soil Index (BSI):

$$BSI = [(B11 + B04) - (B08 + B02)] / [(B11 + B04) + (B08 + B02)]$$

Captures the contrasting spectral slopes between SWIR soil reflectance and VNIR vegetation/water absorption.
- 2. Clay Minerals Ratio (CMR):

$$CMR = B11 / B12$$

Isolates hydroxyl absorption features of aluminosilicate clay minerals (kaolinite, illite, montmorillonite) at 2200 nm.
- 3. Ferrous Iron Index (FII):

$$FII = B12 / B08$$

Measures iron oxide chromophore concentration, highly correlated with tropical soil weathering and phosphorus fixation capacity.
- 4. Carbonate Index (CI):

$$CI = B11 / B08$$

Sensitive to free calcium carbonates (calcite/dolomite), providing vital proxy information for soil pH and alkalinity.
- 5. Normalized Difference Red Edge (NDRE):

$$NDRE = (B08 - B05) / (B08 + B05)$$

Monitors canopy chlorophyll and plant nitrogen uptake, serving as an indirect vegetative proxy for available soil Nitrogen.
- 6. Topographic Wetness Index (TWI):

$$TWI = \ln(a / \tan(\beta))$$

where 'a' represents the specific upslope contributing catchment area and 'beta' denotes the local terrain slope gradient in radians, quantifying hydrological accumulation and soil moisture retention.

6.4 Deep Multi-Modal Learning Backbone

The predictive core of the proposed system leverages a multi-modal hybrid architecture combining a Vision Transformer (ViT) spatial backbone with a cross-attention tabular covariate fusion network:

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- **A. Spatial-Spectral Foundation Backbone:** The spatial-spectral encoder utilizes pre-trained weights from the Clay Foundation Model (ViT-B/16 architecture). Multi-spectral image patches (16 x 16 pixels at 10m resolution, corresponding to 160m x 160m sub-field parcels) are projected into a 768-dimensional latent embedding space via 3D spatio-temporal self-attention, capturing micro-topographic textures and spatial soil color gradients.
 - **B. Cross-Attention Covariate Fusion:** Daily meteorological time-series (temperature, rainfall, soil water tiers from ERA5-Land) and geomorphometric metrics (elevation, slope, TWI) are passed through multi-layer perceptron (MLP) projection heads. A multi-head cross-attention mechanism merges the spatial satellite tokens with the environmental covariate vector, allowing the model to dynamically weight weather and terrain context based on local landscape properties.
 - **C. Heteroscedastic Multi-Task Regressors:** The fused multi-modal representation feeds into five parallel multi-task regression heads corresponding to pH, SOC, Available N, Extractable P, and Exchangeable K. To capture prediction uncertainty, each head outputs both a predicted mean (μ_k) and a log-variance ($s_k = \ln \sigma_k^2$), trained via a Gaussian negative log-likelihood heteroscedastic loss function:

$$\text{Loss} = 0.5 * \sum_{k=1}^5 [\exp(-s_k) * (y_k - \mu_k)^2 + s_k]$$

This formulation penalizes overconfident erroneous predictions while naturally accommodating inherent sensor noise and laboratory measurement error.

7. EXPECTED OUTCOME

7.1 Quantitative Performance Deliverables

The project is engineered to achieve high-precision quantitative predictive capability, outperforming established single-sensor and tabular baselines across all target macro-nutrients and soil health parameters. Target statistical performance benchmarks are detailed in Table 2:

Table 2: Expected Quantitative Model Performance Benchmarks

Soil Property	Benchmark R ² (Kammerlander 2025)	Proposed Target R ²	Target RMSE	Target RPIQ
Soil Organic Carbon (SOC %)	0.58	0.72 - 0.76	$\leq 0.35\%$	≥ 2.10
Available Nitrogen (N kg/ha)	0.42	0.62 - 0.68	≤ 32.0 kg/ha	≥ 1.85
Available Phosphorus (P mg/kg)	0.38	0.56 - 0.62	≤ 4.8 mg/kg	≥ 1.65
Available Potassium (K mg/kg)	0.45	0.60 - 0.66	≤ 24.0 mg/kg	≥ 1.75
Soil pH (Acidity / Alkalinity)	0.64	0.76 - 0.82	≤ 0.42 units	≥ 2.25

7.2 Operational & Socio-Economic Impact

- **Economic Democratization:** Reduces direct testing costs by over 85% compared to commercial wet-chemical laboratories, enabling smallholder farmers to monitor soil health continuously at near-zero marginal cost per acquisition.
- **Real-Time Decision Window:** Compresses analytical turnaround latency from 3-6 weeks to sub-minute automated inference upon satellite image acquisition, enabling timely pre-sowing basal fertilizer decisions.
- **Sub-Field Granularity:** Generates 10m x 10m continuous raster fertility maps (revealing micro-topographic variability) rather than a single coarse point sample per 10 hectares, facilitating true precision agriculture and targeted spot fertilization.
- **Statutory Indian SHC Alignment:** Fully aligns with statutory Indian Soil Health Card (SHC) nutrient classifications (Low, Medium, High), producing direct advisory prescription cards in regional languages for farmer implementation.

7.3 Phase-Wise Implementation Roadmap

- **Phase 1 (V Semester - Completed / Ongoing):** Comprehensive literature review, Copernicus STAC ingestion pipeline development, cloud-masking & bare-soil compositing algorithms, baseline classical ML benchmarking (XGBoost/RF), and academic synopsis defense.
- **Phase 2 (VI Semester - Scheduled):** Geospatial Foundation Model (Clay/Prithvi) fine-tuning, cross-attention tabular covariate integration, heteroscedastic loss implementation, local field validation against certified Karnataka Soil Health Card wet-lab samples, cloud REST API deployment, and final capstone thesis presentation.

8. REFERENCES

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